



Application of deep learning models for accurate classification of fluid collections in acute necrotizing pancreatitis on computed tomography: a multicenter study

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Abstract

Purpose To apply CT-based deep learning (DL) models for accurate solid debris-based classification of pancreatic fluid collections (PFC) in acute pancreatitis (AP).

Material and methods This retrospective study comprised four tertiary care hospitals. Consecutive patients with AP and PFCs who had computed tomography (CT) prior to drainage were screened. Those who had magnetic resonance imaging (MRI) or endoscopic ultrasound (EUS) within 20 days of CT were considered for inclusion. Axial CT images were utilized for model training. Images were labelled as those with $\leq 30\%$ solid debris and $>30\%$ solid debris based on MRI or EUS. Single center data was used for model training and validation. Data from other three centers comprised the held out external test cohort. We experimented with ResNet 50, Vision transformer (ViT), and MedViT architectures.

Results Overall, we recruited 152 patients (129 training/validation and 23 testing). There were 1334, 334 and 512 images in the training, validation, and test cohorts, respectively. In the overall training and validation cohorts, ViT and MedViT models had high diagnostic performance (sensitivity 92.4–98.7%, specificity 89.7–98.4%, and AUC 0.908–0.980). The sensitivity (85.3–98.6%), specificity (69.4–99.4%), and AUC (0.779–0.984) of all the models was high in all the subgroups in the training and validation cohorts. In the overall external test cohort, MedViT had the best diagnostic performance (sensitivity 75.2%, specificity 75.3%, and AUC 0.753). MedViT had sensitivity, specificity, and AUC of 75.2%, 74.3%, and 0.748, in walled off necrosis and 79%, 74.2%, 75.3%, and 0.767 for collections >5 cm.

Conclusion DL-models have moderate diagnostic performance for solid-debris based classification of WON and collections greater than 5 cm on CT.

Keywords Acute necrotizing pancreatitis · Computed tomography · Deep learning

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Introduction

Acute pancreatitis (AP) is a common condition presenting to the emergency department. Moderately severe and severe AP are associated with systemic and local complications that cause significant morbidity and mortality [1]. Pancreatic fluid collection (PFC) is an important local complication of AP that leads to significant utilization of healthcare resources for its management [2]. The standard strategy for managing infected or symptomatic PFC is the step-up approach [3]. The step-up approach initially involves endoscopic drainage (ED) or percutaneous drainage [3, 4]. Patients who fail to respond to drainage alone undergo endoscopic necrosectomy (EN) or surgical necrosectomy [4, 5].

Imaging is critical in guiding the treatment strategy for PFCs. Contrast enhanced computed tomography (CECT) is the imaging technique of choice for assessing patients with AP [1]. The extent of a PFC is well delineated by CECT [6]. However, CECT is poor in evaluating the solid component within a PFC which is a key factor influencing the technique of drainage/ necrosectomy [7–10]. Patients with extensive solid debris need lumen apposing metal stents and repeated interventions and have poorer outcomes than those with predominantly liquid collections [10, 11]. Solid debris are adequately visualised on magnetic resonance imaging (MRI) and endoscopic ultrasound (EUS) [7–9]. In a study comparing CT and MRI for detecting solid debris and determining the drainability of fluid collection, CT had 25% sensitivity compared with 100% sensitivity for MRI [8]. In another study, solid debris was detected in 92% patients with EUS compared to 32% patients with CT ($p < 0.001$) [9]. Another study reported that CT had a lower reproducibility for fluid to debris component estimation in PFCs [10]. However, routine use of MRI/ EUS for characterizing fluid collections is not feasible due to limited availability and high cost (MRI) and invasiveness and patients discomfort (EUS). Additionally, EUS cannot accurately evaluate collections in the deeper planes (pararenal spaces and paracolic gutters) and there is a poor agreement between the endosonologists for the percentage of solid debris [12]. Methods to enhance the ability to detect and quantify solid debris on CT are highly desirable.

Deep learning (DL)-based algorithms can identify hidden patterns in medical images that can improve the classification tasks [13]. Machine learning (ML) models have shown to improve the prediction of severity of AP [14–17]. However, to our knowledge, there are no DL models to classify PFCs based on the solid component on CECT images.

We aim to apply DL models to accurately classify PFCs based on the amount of solid debris on CT images using MRI/EUS images as the reference standards.

Methods

Study design and participants

This retrospective multicentre study comprised four tertiary care referral hospitals. The study was approved by the institutional ethics committees and the need for informed written consent was waived. Consecutive patients with acute necrotizing pancreatitis and PFCs who had CT prior to drainage were screened. Those who had the MRI/EUS evaluation within 2 weeks of CT were considered for inclusion. Patients with history of drainage procedures prior to CT or in the interval between CT and MRI/EUS were excluded. The diagnosis of AP was based on Revised Atlanta Classification (RAC) [18]. The clinical details including age, gender, cause of AP, RAC severity of AP, type of PFC [acute necrotic collection (ANP) or walled off necrosis (WON), per RAC], and interval between a CT and MRI/ EUS were recorded.

CT technique

All CT scans were performed on multidetector row CT scanners following intravenous injection of 80–100 ml of non-ionic iodinated contrast agent using a pressure injector at a rate of 2.5–3.5 mL/second. The CT acquisition was either a single portal venous phase (70–90 s) or biphasic comprising arterial and portal venous phase. Biphasic CT scans were performed in patients with suspected bleeding complications.

MRI technique

MRI scans were performed across all the centers using 1.5-T or 3-T MRI scanners. At all the centers, the minimum protocol for MRI was axial and coronal half Fourier single shot turbo spin echo (HASTE) images, axial T1-weighted images, and magnetic resonance cholangiopancreatography images.

Data preparation and annotation

For the training of models, only portal venous phase images were utilized.

Axial portal venous phase CT images of slice thickness 3–5 mm were selected by a radiologist with 10 years of experience in abdominal imaging. Images were anonymized and saved in jpeg format by research fellows before feeding them to the DL algorithm. The input to the DL algorithm

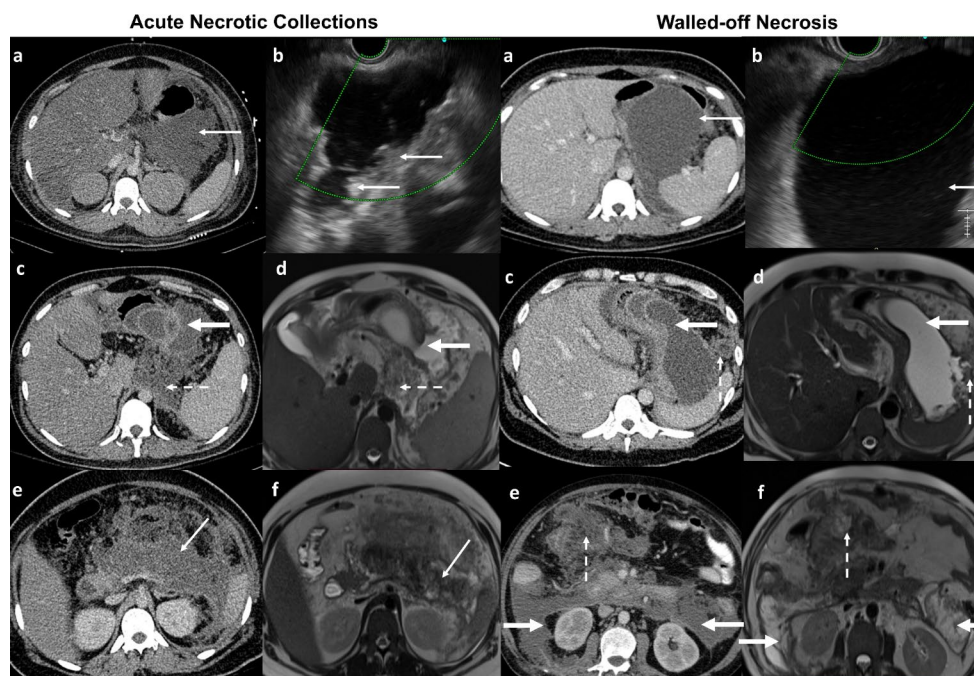


Fig. 1 Representative images from the training dataset. Acute necrotic collections. **a** and **b**. Contrast enhanced CT and EUS images of a 54-year-old male with ethanol related acute pancreatitis in the third week of illness (day 19) show a collection in gastrohepatic location (arrow, **a**) with small amount ($<30\%$) solid debris on EUS image (**b**). **c** and **d**. CT and MRI images of a 41-year-old male with ethanol related acute pancreatitis in the third week of illness (day 17) show a lesser sac collection. The posterior component of the collection (arrow, **c**) comprises predominantly of solid debris (dashed arrow, **d**). The anterior component (arrow, **c**) is predominantly liquid (arrow, **d**). Overall, the collection has more than 30% solid debris. **e** and **f**. CT and MRI images of a 37-year-old female with gallstone related acute pancre-

atitis show lesser sac collection (arrow, **e**). In MRI, the collection is comprising predominantly of solid necrotic pancreatic tissue (arrow, **f**). Walled-off necrosis. **A** and **B**. CT and EUS images of a 23-year-old male with acute pancreatitis of unknown etiology show a collection (arrow, **a**) without any solid debris (arrow, **b**). **c** and **d**. CT and MRI images of a 53-year-old female with gallstone related acute pancreatitis show a collection in lesser sac (arrows, **c** and **d**) comprising with $<30\%$ solid debris (dashed arrows, **c** and **d**). **e** and **f**. CT and MRI images in a 66-year-old male with alcohol related acute pancreatitis show collections in bilateral pararenal spaces (thick arrows, **e**) showing $<30\%$ solid debris (arrows, **f**). The omental collection (dashed arrow, **e**) is predominantly solid (dashed arrow, **f**)

was the entire CT image. No region of interest was drawn before inputting the images to the DL algorithm.

While the amount of solid debris is critical in the choice of stents, needs for necrosectomy, and clinical outcomes, a best single cut-off is lacking [19]. We chose a cut-off of 30% solid debris to train the models based on the practice at our center (placement of lumen apposing metallic stents for collection with $>30\%$ solid debris and plastic stents for collections with $\leq 30\%$ solid debris and higher frequency of necrosectomy in the group with $>30\%$ solid debris). A recent publication proposed the stratification of walled off necrosis based on the extent of necrosis, amount of solid debris, and infection [20]. The authors reported poor outcomes in patients with $>30\%$ solid debris. The image labels for PFCs, viz. $\leq 30\%$ solid debris or $>30\%$ solid debris were based on the image-to-image comparison of CT and MRI images by a radiologist with 10 years of post-training experience in abdominal MRI. For the class labels, we read axial HASTE images only. In cases where the EUS-based reference standard was utilized, the image labels were based on the reporting of the solid component by expert gastroenterologists

with 2–5 years of post-training experience in EUS. The case where abdominal MRI was available, the CT images of the entire extent of the collection were labelled by expert radiologists and utilized for training while for EUS-based labels, only upper abdominal CT images were selected and labelled by expert radiologists and gastroenterologists in consensus. Figure 1 shows the representative images from the dataset.

Training, validation, and testing dataset

Training and validation data

The training data was acquired by a single center between January 2020 and June 2023 (*internal dataset*). Validation data comprised 20% of the training data created randomly using a pytorch program.

Testing data

Three centers contributed the test dataset (*external dataset*). These centres collected data retrospectively between

January 2023 and October 2023. This dataset was completely held out and not used for training.

Supplementary Table 1 shows the distribution of data from different centres.

Neural network implementation

We experimented with ResNet 50 (<https://github.com/pytorch/vision/blob/main/torchvision/models/resnet.py>), Vision transformer (ViT) (https://github.com/pytorch/vision/blob/main/torchvision/models/vision_transformer.py), and Med ViT (<https://github.com/Omid-Nejati/MedViT>) representing convolutional neural network (CNN)-based, transformer-based, and hybrid architectures for training, respectively [21–23]. ResNet 50 is the most popular CNN image classification network. ResNet-50 has residual connections to the CNN layers [21]. This architecture maintains expressive power while reducing the computational complexity. Transformers, on the other hand, apply the architecture directly to image data without relying on convolutional layers and processing the image as a sequence of patches. Typically, ViT processes images as sequences of non-overlapping patches of fixed size, and also utilizes self-attention mechanisms to capture long-range dependencies [22]. MedViT is an efficient CNN-Transformer hybrid model with the locality of CNNs as well as the global connectivity of ViT, pre-trained on medical images [23].

Training the neural networks

The weights of the various neural networks were initialised from the pretrained models. We modified the classification head to binary classification. We used the binary cross-entropy loss (BCE) as the objective function. We trained the networks end-to-end with a stochastic gradient descent (ViT and MedViT) or adaptive moment estimation optimizer (ResNet 50) with an initial learning rate of 0.001 for ResNet 50 and MedViT and 0.0008 for ViT, a momentum of 0.9, and a weight decay of 0.0005. The mini-batch size was 32 for ResNet 50 and ViT and 4 for MedViT. We used the resize, random crop, random horizontal flip with probability 0.1, and normalisation for data augmentations during the training to mitigate overfitting issues. The input image size to all the networks was 224×224 . We trained the ResNet 50 for 100 epochs, ViT for 60 epochs and MedViT for 40 epochs. We used a validation set of 20% of the training data selected randomly by a pytorch program to optimize the network weights and the hyperparameters. The best-performing model on the validation set in terms of accuracy was selected for evaluation on a held-out test set. The models were trained and evaluated with the Tesla K80 GPU.

Statistical analysis

To evaluate the classification results of the DL models for the overall group and the various subgroups (ANC vs. WON, size < 5 cm vs. size > 5 cm), we used sensitivity, specificity, accuracy, positive predictive value, negative predictive value, and area under the receiver operating characteristic curve (AUC). Diagnostic performance measures were calculated using the mean and 95% confidence intervals. The sensitivity and specificity were compared using McNemar test and AUC was compared using Delong method. The statistical analyses were performed using IBM® SPSS® version 22 (IBM India Pvt Ltd, India), MedCal® version 19.4 (MedCalc Software Ltd, Belgium), and Analyse-it® version 6.15 (Analyse-it Software Ltd. UK).

Results

Patients

Overall, we evaluated 2180 CT images of 152 patients. There were 129 and 23 patients in training and testing cohorts, respectively (Fig. 2). The demographic characteristics of the various cohorts is given in Table 1. There were 1334, 334 and 512 images in training, validation, and testing cohorts, respectively. MRI and EUS were available as reference standard in 67 and 85 patients, respectively.

Diagnostics performance of DL models

Overall group

In the training cohort, ViT and MedViT had similar diagnostic performance. The sensitivity, specificity, PPV, NPV, and AUC of ResNet50 was significantly lower than ViT and MedViT models (Table 2). In the validation cohort, MedViT had the significantly better diagnostic performance (sensitivity, specificity, and AUC of 95.4%, 91.6%, and 93.6%, respectively) than ResNet 50 and ViT (Table 3).

In the external test cohort, MedViT had the best diagnostic performance with sensitivity, specificity, accuracy, and AUC of 75.2%, 75.3%, 75.3% and 0.753, respectively. The AUC of MedViT was significantly greater than ViT and ResNet 50 (Table 4).

Figure 3 shows the ROC curves of the overall group in the training, validation, and test cohorts.

Subgroup analysis

In general, the sensitivity, specificity, accuracy, and AUC of all the models was high in all the subgroups in the training

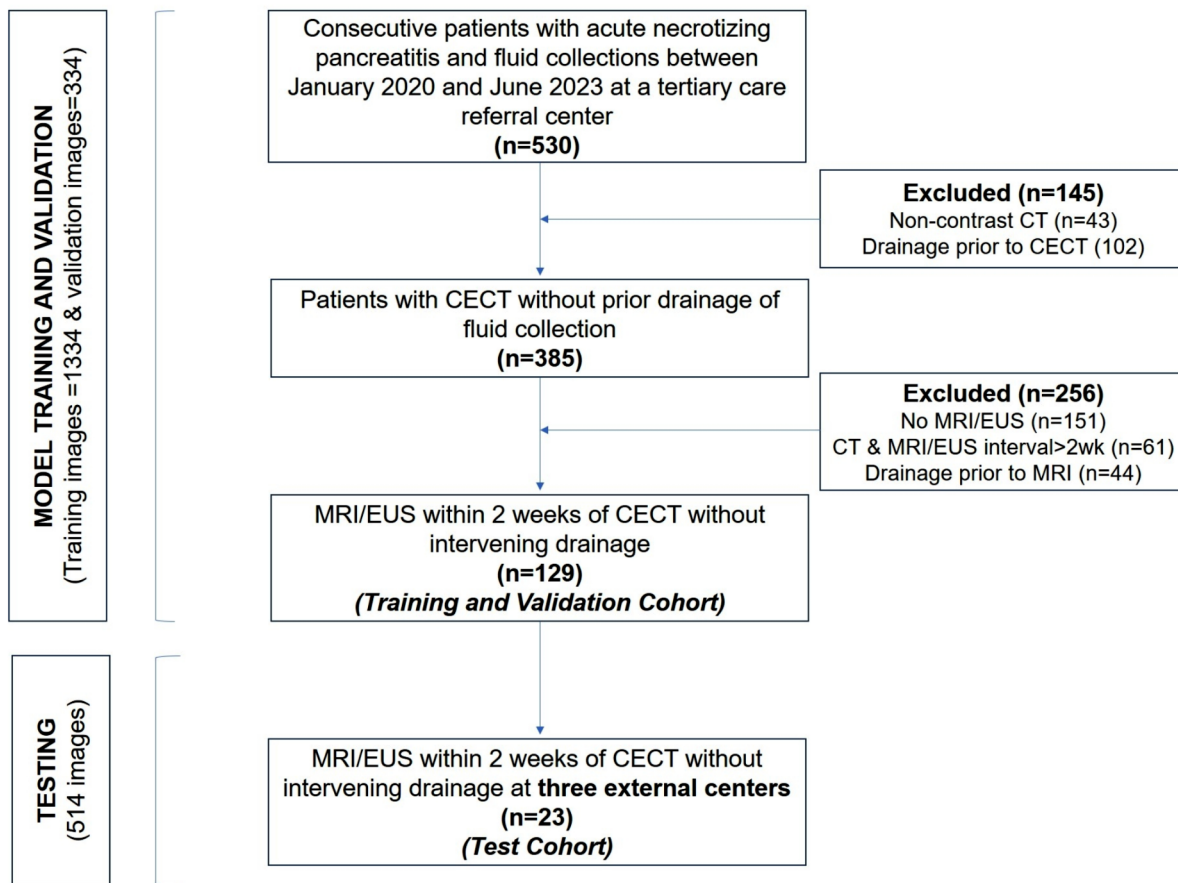


Fig. 2 Flow diagram showing the recruitment of training, validation, and test cohorts. One center contributed the training and validation data. Testing was done on held out external dataset from three different centers

Table 1 Demographic characteristics of the study population

Parameters	Training and validation cohort (n = 129)	Test cohort (n = 23)
Mean age (±SD)	36.8±11.63 years	36.4±12.7 years
Female/ Male	32(24.8%)/97(75.2%)	03(13%)/20(87%)
Severity of Pancreatitis	50 (38.7%)	08 (34.8%)
-Moderately severe	79 (61.3%)	15 (65.2%)
-Severe		
Cause of Pancreatitis	59 (45.7%)	08 (34.8%)
-Gallstone disease	54 (41.9%)	10 (43.5%)
-Ethanol	12 (9.3%)	03 (13%)
-Other causes	04 (3.1%)	02 (8.7%)
-Idiopathic		
Mean interval between CT and MRI/EUS (±SD)	7.8 (±4.9) days	7.2 (±5.6) days
Type of collection	43 (33.3%)	07 (30.4%)
-ANC	86 (67.7%)	16 (69.6%)
-WON		
Mean size of collection (±SD)	10.4±4.4 cm	11.4±3.5 cm
Classification based on solid debris	71 (55%)	15 (65.2%)
-≤30%	58 (45%)	08 (34.8%)
->30%		

SD: standard deviation

and validation cohorts. In the test cohort, the diagnostic performance of all the models was high for WON and size > 5 cm. MedVit had sensitivity, specificity, accuracy, and AUC of 75.2%, 74.3%, 74.5% and 0.748, respectively for WON and 79%, 74.2%, 75.3%, and 0.767, respectively for size > 5 cm in the test cohort. The diagnostic performance was relatively poor for ANC and smaller collections in the test cohort. Tables 2, 3 and 4 shows the diagnostic performance in various subgroups. Table 5 shows the p-values of comparison of various models across the cohorts.

Figure 4 shows the class activation maps. Supplementary Figs. 1–9 show the AUCs for the overall group and various subgroups in the training, validation, and test cohorts.

Discussion

In this multicenter study evaluating DL models for classification of PFCs based on the amount of solid debris, we found that all the CT-based DL models achieved high diagnostic performance in the internal training and validation cohorts. In the external unseen test cohort also, DL models showed moderate diagnostic performance. On subgroup analysis, best diagnostic performance was achieved for

Table 2 Diagnostic performance of deep learning models in training cohort

Group	Model	Sens. (95% CI)	Spec. (95% CI)	Acc. (95% CI)	PPV (95% CI)	NPV (95% CI)	AUC (95% CI)
Overall	ResNet50	88.8 (86.3–91.1)	81.4 (78.3–84.2)	85.1 (83.1–86.9)	82.4 (8.8–84.6)	88.1 (85.7–9.02)	0.849 (0.822–0.868)
	ViT	98.7 (97.5–99.4)	97.3 (95.8–98.3)	98.0 (97.1–98.6)	97.3 (95.8–98.2)	98.7 (97.5–99.3)	0.979 (0.971–0.986)
	MedVit	97.6 (96.2–98.6)	98.4 (97.2–99.2)	98.0 (97.2–98.7)	98.4 (97.1–99.1)	97.7 (96.4–98.6)	0.980 (0.971–0.987)
ANC	ResNet50	85.6 (80.7–89.6)	99.2 (96.0–99.9)	90.3 (87.0–93.1)	99 (96.89–99.9)	78.74% (73.3–83.3)	0.783 (0.744–0.821)
	ViT	98.8 (96.6–99.7)	97.9 (94.7–99.4)	98.4 (96.8–99.3)	98.4 (96.0–99.4)	98.4 (95.3–99.4)	0.984 (0.971–0.995)
	MedVit	96.5 (93.4–98.3)	99.4 (97.1–99.9)	97.7 (95.9–98.9)	99.6 (97.2–99.9)	95.5 (91.8–97.5)	0.980 (0.967–0.992)
WON	ResNet50	89.6 (86.1–92.5)	85.8 (82.5–88.7)	87.4 (85.1–89.5)	82.4 (79.0–85.3)	91.8 (89.2–93.7)	0.878 (0.856–0.899)
	ViT	98.4 (96.5–99.4)	97.0 (95.1–98.3)	97.6 (96.4–98.5)	96.1 (93.7–97.6)	98.8 (97.3–99.4)	0.977 (0.967–0.987)
	MedVit	98.1 (96.2–99.2)	98.0 (96.4–99.0)	98.0 (96.9–98.8)	97.3 (95.2–98.5)	98.6 (97.1–99.3)	0.981 (0.971–0.990)
Less than 5 cm	ResNet50	90.8 (86.1–94.3)	75.7 (69.9–81.2)	83.1 (79.3–86.5)	78.2 (74.0–81.9)	89.5 (84.9–92.9)	0.833 (0.799–0.866)
	ViT	98.6 (96.0–99.7)	94.7 (90.9–97.2)	96.6 (94.5–98.1)	94.7 (91.1–96.8)	98.6 (95.8–99.5)	0.967 (0.950–0.983)
	MedVit	97.8 (94.9–99.2)	97.7 (94.7–99.2)	97.7 (95.9–98.9)	97.8 (94.9–99.0)	97.7 (94.7–99.0)	0.978 (0.963–0.991)
More than 5 cm	ResNet50	86.5 (82.8–89.6)	84.2 (80.6–87.4)	85.3 (82.8–87.5)	82.7 (79.5–85.5)	87.7 (84.8–90.1)	0.856 (0.832–0.878)
	ViT	98.5 (96.8–99.4)	98.5 (96.9–99.4)	98.5 (97.5–99.2)	98.3 (96.5–99.1)	98.7 (97.2–99.4)	0.985 (0.977–0.993)
	MedVit	97.3 (95.3–98.6)	98.7 (97.2–99.5)	98.0 (96.9–98.8)	98.5 (96.8–99.3)	97.7 (95.9–98.7)	0.980 (0.977–0.989)

ANC: acute necrotic collection, AUC: area under the curve, CI: confidence interval; NPV: negative predictive value, PPV: positive predictive value, WON: walled off necrosis

Table 3 Diagnostic performance of deep learning models in validation cohort

Group	Model	Sens. (95% CI)	Spec. (95% CI)	Acc. (95% CI)	PPV (95% CI)	NPV (95% CI)	AUC (95% CI)
Overall	ResNet50	87.8 (82.0–92.3)	82.3 (75.9–87.7)	85.1 (80.9–88.6)	83.0 (78.0–87.1)	87.3 (82.1–91.2)	0.845 (0.806–0.884)
	ViT	92.4 (87.4–95.9)	89.7 (83.4–95.9)	91.1 (87.6–93.8)	89.8 (85.1–93.2)	92.4 (87.6–93.8)	0.908 (0.876–0.938)
	MedVit	93.4 (88.9–96.7)	92.6 (87.7–96.0)	93.1 (89.9–95.5)	92.5 (88.0–95.4)	93.6 (89.3–96.3)	0.928 (0.900–0.956)
ANC	ResNet50	86.3 (75.6–93.5)	69.4 (51.8–83.6)	80.3 (71.3–87.5)	83.8 (75.8–89.5)	73.5 (59.3–84.1)	0.779 (0.692–0.866)
	ViT	90.9 (81.2–96.5)	75.0 (57.8–87.8)	85.2 (76.9–91.5)	86.9 (79.0–92.1)	81.2 (67.2–90.8)	0.830 (0.749–0.909)
	MedVit	95.4 (87.2–99.0)	91.6 (77.5–98.2)	94.1 (87.6–97.8)	95.4 (87.6–98.4)	91.6 (78.3–97.0)	0.936 (0.883–0.987)
WON	ResNet50	86.9 (78.3–93.0)	85.7 (78.8–91.0)	86.2 (81.0–90.3)	80.0 (72.5–85.8)	90.9 (85.4–94.4)	0.863 (0.818–0.908)
	ViT	92.3 (84.9–96.8)	93.5 (88.1–97.0)	93.1 (89.0–96.0)	90.4 (83.3–94.6)	94.9 (90.1–97.4)	0.930 (0.895–0.963)
	MedVit	91.3 (83.5–96.1)	92.8 (87.2–96.5)	92.2 (88.0–95.3)	89.3 (82.1–93.8)	94.2 (89.3–96.9)	0.921 (0.884–0.956)
Less than 5 cm	ResNet50	91.4 (76.9–98.2)	92.8 (76.5–99.1)	92.0 (82.4–97.3)	94.1 (80.7–98.3)	89.6 (74.5–96.2)	0.921 (0.853–0.989)
	ViT	94.2 (80.8–99.3)	92.8 (76.5–99.1)	93.6 (84.5–98.2)	94.2 (81.2–98.4)	92.8 (77.1–98.0)	0.936 (0.873–0.998)
	MedVit	97.1 (85.0–99.9)	96.4 (81.6–99.9)	96.8 (89.0–99.6)	97.1 (83.2–99.5)	96.4 (79.6–99.4)	0.968 (0.923–1.00)
More than 5 cm	ResNet50	85.3 (77.8–91.0)	80.4 (73.0–86.4)	82.6 (77.6–86.9)	78.3 (72.1–83.5)	86.8 (81.0–91.0)	0.829 (0.784–0.873)
	ViT	91.0 (84.5–95.5)	89.1 (83.0–93.6)	90.0 (85.8–93.3)	87.1 (81.4–91.7)	92.3 (87.1–95.4)	0.901 (0.865–0.936)
	MedVit	91.8 (85.5–96.0)	91.8 (86.2–95.7)	91.8 (87.9–94.8)	90.4 (84.5–94.2)	93.1 (88.2–94.8)	0.919 (0.886–0.951)

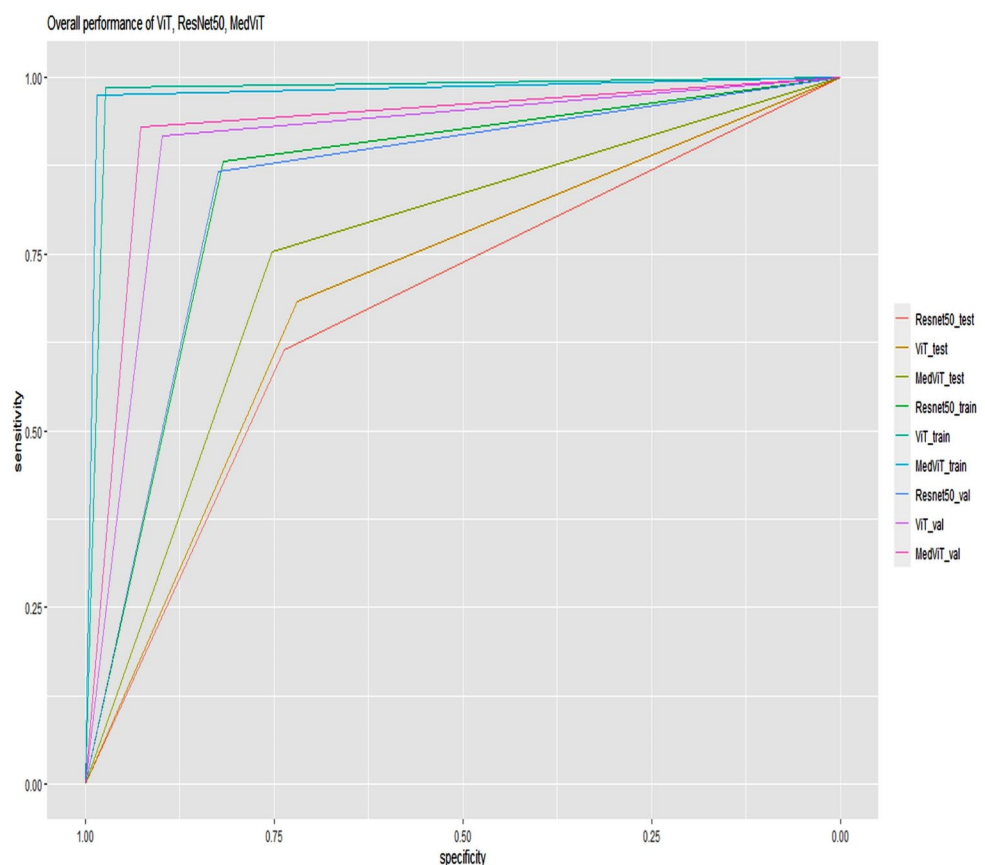
ANC: acute necrotic collection, AUC: area under the curve, CI: confidence interval; NPV: negative predictive value, PPV: positive predictive value, WON: walled off necrosis

Table 4 Diagnostic performance of deep learning models in test cohort

Group	Model	Sens. (95% CI)	Spec. (95% CI)	Acc. (95% CI)	PPV (95% CI)	NPV (95% CI)	AUC (95% CI)
Overall (512)*	ResNet50	61.3 (51.1–70.9)	73.6 (69–77.8)	71.2 (67–75)	36.2 (31.2–41.5)	88.6 (85.8–90.9)	0.675 (0.622–0.727)
	ViT	68.3 (67–75.9)	71.6 (67–75.9)	71 (66.8–74.9)	37.1 (32.5–41.9)	90.2 (87.3–92.5)	0.701 (0.650–0.751)
	MedVit	75.2 (65.6–83.3)	75.3 (70.8–79.3)	75.3 (71.3–78.9)	42.7 (37.8–47.7)	92.5 (89.8–94.6)	0.753 (0.705–0.799)
ANC (101)	ResNet50	-	84.1 (75.5–90.6)	-	-	100 (95.7–100)	-
	ViT	-	79.2 (69.9–86.6)	-	-	100 (95.4–100)	-
	MedVit	-	78.2 (68.9–85.8)	-	-	100 (95.4–100)	-
WON (413)	ResNet50	61.3 (51.1–70.9)	77.4 (73.1–81.4)	74.3 (70.3–78)	40 (34.4–45.7)	89.1 (86.4–91.3)	0.658 (0.603–0.712)
	ViT	68.3 (58.3–77.2)	69.2 (63.7–74.3)	69 (64.3–73.4)	41.8 (36.7–47)	87.1 (83.3–90)	0.688 (0.635–0.740)
	MedVit	75.2 (65.6–83.3)	74.3 (69.1–79.1)	74.5 (70–78.7)	48.7 (43.3–52.2)	90.2 (86.7–92.9)	0.748 (0.699–0.796)
Less than 5 cm (117)	ResNet50	6.6 (0.1–31.9)	83.3 (74.6–89.9)	73.5 (64.5–81.2)	5.5 (0.8–29.1)	85.8 (83.7–87.7)	0.450 (0.375–0.524)
	ViT	26.6 (7.7–55.1)	78.4 (69.1–85.9)	71.7 (62.7–79.7)	15.3 (6.7–31.2)	87.9 (84.0–90.9)	0.530 (0.408–0.652)
	MedVit	53.3 (26.5–78.7)	81.2 (72.9–87.8)	78 (70–84.7)	26.6 (16.5–39.9)	93.1 (88.7–95.9)	0.659 (0.522–0.795)
More than 5 cm (397)	ResNet50	70.9 (60.1–80.2)	70.4 (65–80.2)	70.5 (65.7–74.9)	39.8 (34.7–45.2)	89.7 (86.2–92.4)	0.707 (0.652–0.761)
	ViT	75.5 (65.1–84.2)	69.4 (64–74.5)	70.7 (66–75.2)	40.6 (35.7–45.6)	91.1 (87.5–93.7)	0.725 (0.672–0.777)
	MedVit	79 (68.9–87.1)	74.2 (69–79)	75.3 (70.7–79.4)	45.9 (40.6–51.3)	92.7 (89.4–95.1)	0.767 (0.717–0.816)

*Number of images; ANC: acute necrotic collection, AUC: area under the curve, CI: confidence interval; NPV: negative predictive value, PPV: positive predictive value, WON: walled off necrosis

Fig. 3 Area under curve in the overall group. Area under the receiver operating characteristic curve of the three deep learning models for the detection of >30% solid debris in the overall training, validation, and test cohorts



WON and collections larger than 5 cm in the test cohort. These results suggest that DL-based classification of fluid collections may be used in clinical practice.

The amount of solid debris is critical in guiding the management decisions in PFCs [10, 11, 19, 20]. Accurate depiction of solid debris within PFCs on CT is challenging [7–9].

Although, EUS and MRI are accurate in depicting solid debris in PFCs, their use in clinical practice is challenging due to availability, cost, or invasiveness [7–9]. As CT is the most commonly used and widely available, methods to improve its performance in detection of solid debris is desirable. AI has shown promise in various medical imaging

Table 5 p-values for comparison of models across the cohorts

Group	Comparison	p-value								
		Train			Validation			Test		
		ResNet vs. ViT	ResNet Vs. MedViT	ViT vs. MedViT	ResNet vs. ViT	ResNet Vs. MedViT	ViT vs. MedViT	ResNet vs. ViT	ResNet Vs. MedViT	ViT vs. MedViT
Overall	Sensitivity	<0.0001	<0.0001	0.052	0.091	0.025	0.414	0.161	0.006	0.144
	Specificity	<0.0001	<0.0001	0.032	0.012	0.007	0.095	0.458	0.540	0.212
	AUC	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
ANC	Sensitivity	<0.0001	<0.0001	0.014	0.256	0.033	0.083	-	-	-
	Specificity	<0.0001	<0.0001	0.083	0.563	0.011	0.0143	0.317	0.157	0.593
	AUC	-	-	-	-	-	-	-	-	-
WON	Sensitivity	<0.0001	<0.0001	0.705	0.131	0.248	0.563	0.006	0.006	0.001
	Specificity	<0.0001	<0.0001	0.131	0.004	0.018	0.563	0.0011	0.221	0.001
	AUC	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	-	-	-
Less than 5 cm	Sensitivity	<0.0001	0.0006	0.157	0.563	0.313	0.313	0.083	0.008	0.102
	Specificity	<0.0001	<0.0001	0.008	1.000	0.313	0.313	0.317	0.251	0.796
	AUC	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.626	0.022	0.189
More than 5 cm	Sensitivity	<0.0001	<0.0001	0.131	0.070	0.045	0.654	0.393	0.108	0.466
	Specificity	<0.0001	<0.0001	0.705	0.009	0.001	0.517	0.725	0.256	0.154
	AUC	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001

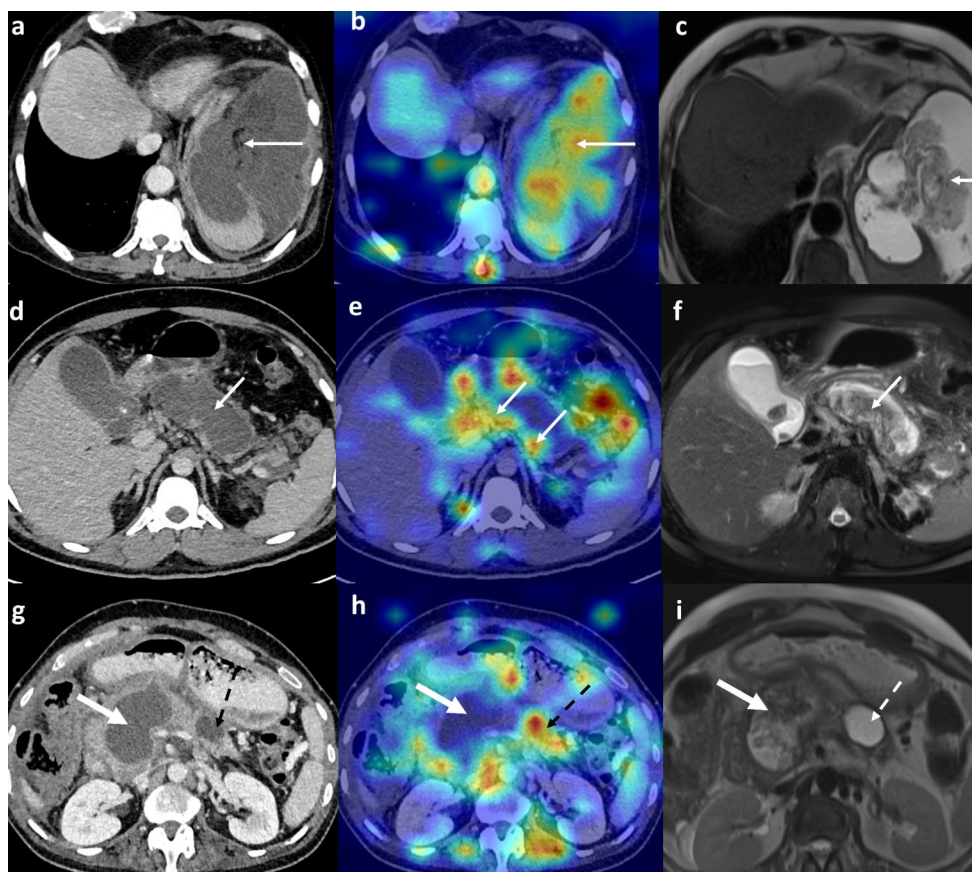


Fig. 4 Gradient class activation maps (GradCAM). Heat maps are plotted to see whether the model predictions are based on learning or whether the model makes random predictions. These are not used to assess the correctness of classification. a–c. CT image (a) shows a wall-off necrotic collection in the gastrosplenic region (arrow). The deep learning (DL) model accurately predicted the collection as solid. GradCAM (b) shows that the model attended to the correct regions within the collection to classify the collection as solid. MRI image (c) shows >30% solid debris within the collection (arrow). d–f. CT image shows a collection in the

lesser sac (arrow, d). The DL model correctly predicted the collection as solid (>30% solid debris) based on attention (arrow, e) on the solid debris (arrow, f). g–i. CT image shows collections in the pancreas (arrow and thick arrow, g). The DL model incorrectly predicted the collection as liquid based on attention (dashed arrow, h) on the smaller liquid component (dashed arrow, i). However, the major portion of the collection is solid (arrow, g and i). These examples highlight that model predictions were based on learning and were not random

tasks. In image classification, AI has shown accuracy comparable to expert radiologist. Several studies have reported the potential of AI in predicting the severity of AP and guiding management. A recent study reported an AUC of 0.757 of a machine learning (ML) XGBoost classifier for pancreatic necrosis prediction in 2387 patients [14]. Another recent study reported a high sensitivity and specificity of ML algorithms in predicting the severity of AP in a cohort of 1012 patients [15]. An EASY APP has been developed as a part of a prospective multicenter study for accurately identifying patients at high risk of severe AP within a first few hours of hospitalization [16]. A systematic review identified 24 articles comprising 8,327 patients and 47 models in AP. Prediction of severity, complication, mortality, recurrence and timing of surgery was reported in 10, 10, 3, 3, and 2 studies, respectively [17].

However, to our knowledge, none of the studies reported using AI for detecting solid debris on CT images. We used three different DL architectures to train models. We employed ResNet 50, ViT, and MedVit that represent CNN, transformer, and hybrid (CNN+transformer), respectively. We found that all the DL-models achieved high diagnostic performance in the overall group and various subgroups in the training and validation cohorts. In the held out external dataset (from three different centers), there was moderate performance in the overall group, WON, and collection greater than 5 cm. However, the performance was poor for ANC and collections smaller than 5 cm. The relatively poor performance for ANC and smaller collections may be due to the smaller training data for these subgroups. Additionally, feeding candidate region proposals (using architectures like Faster RCNN) rather than the entire image for training the classification models may have achieved a better diagnostic performance in the test cohort [24]. Finally, it is well known that the generic classification architectures lack domain adaptation and domain generalization. Thus, future works must incorporate novel strategies to achieve domain adaptation and domain generalization to achieve high diagnostic performance in the unseen data.

There are several limitations to our study. First, the training data was small. Second, the classification of liquid ($\leq 30\%$ solid debris) vs. solid ($> 30\%$ solid debris) classes was based on the practice followed at our center. However, there is no guideline or consensus on this classification. Third, the true labels for training and test data comprised MRI or EUS. A uniform reference standard of MRI would have been ideal. Fourth, the true labels were based on MRI/EUS evaluation by a single experienced radiologist/ gastroenterologist. Fifth, the interval between CT and MRI/ EUS of up to 2 weeks may longer. Ideally, both the investigations should be done on the same day or within a period of less than 3 days. However, this is not feasible in a retrospective

dataset. Sixth, there were limited images of ANC and collections less than 5 cm, which may have led to poor performance of DL-models for classification of these subgroups in the test set. Finally, we did not evaluate the impact of DL-based classification in guiding drainage.

In conclusion, our study shows that DL models have a moderate diagnostic performance for CT-based classification of solid debris in in WON and fluid collections greater than 5 cm. However, larger multicenter datasets need to be curated to further fine tune these models.

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Declarations

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Conflict of interest None.

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